

Residual Asset Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Residual Asset Burden (RAB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on RAB achieves an annualized gross (net) Sharpe ratio of 0.39 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (22) bps/month with a t-statistic of 3.58 (3.24), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Statman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple) is 21 bps/month with a t-statistic of 3.02.

1 Introduction

Financial markets efficiently incorporate information about firms’ productive capacity and investment opportunities into asset prices, yet substantial cross-sectional variation in expected returns persists even after controlling for standard risk factors. This variation reflects differences in firms’ exposure to aggregate productivity shocks and their ability to adjust capital stock in response to changing economic conditions. We examine whether Residual Asset Burden (RAB), a measure that captures the friction-adjusted cost of maintaining and deploying productive assets, predicts cross-sectional differences in stock returns through a production-based asset pricing lens. The persistence of return predictability associated with asset-based measures suggests that markets price the real frictions firms face when transforming inputs into outputs, particularly when these frictions interact with time-varying investment opportunities [Cochrane \(1991\)](#); [Zhang \(2005a\)](#).

In a production-based economy, firms facing higher marginal costs of adjusting their capital stock require higher expected returns to compensate investors for bearing systematic risk associated with productivity shocks [Cochrane \(1991\)](#). Residual Asset Burden captures the component of asset intensity that cannot be explained by standard operating characteristics, thereby isolating the friction-specific burden that assets impose on firms’ production functions. When firms carry high RAB, they face elevated exposure to aggregate productivity shocks because their inflexible asset base amplifies the impact of demand fluctuations on profitability [Zhang \(2005a\)](#); [Liu et al. \(2009\)](#). This mechanism operates through the q-theory channel: firms with high adjustment costs optimally choose lower investment rates, which correspond to higher discount rates in equilibrium [Liu et al. \(2009\)](#).

The production function provides the theoretical foundation linking RAB to expected returns through conditional risk premia. Firms with high residual asset burdens operate with production technologies characterized by decreasing returns to

scale when adjustment costs bind, making their cash flows particularly sensitive to aggregate productivity innovations [Balvers and Zhang \(2017\)](#). These firms cannot easily redeploy assets during negative productivity shocks, creating asymmetric exposure to systematic risk that investors demand compensation for bearing. The convexity of adjustment costs implies that this risk premium increases non-linearly with the degree of asset burden, as marginal adjustment costs rise faster than average costs [Cooper and Haltiwanger \(2006\)](#).

Furthermore, RAB proxies for firms' exposure to investment-specific technology shocks, which constitute a priced source of systematic risk in production economies [Papanikolaou \(2011\)](#). High RAB firms exhibit limited operational flexibility to adopt new technologies or reconfigure their productive capacity in response to technological innovations. This technological inflexibility manifests as higher systematic risk because these firms' values decline disproportionately when positive technology shocks benefit their more flexible competitors. The interaction between asset burden and technology shocks generates time-varying risk premia that standard factor models fail to capture fully, as these models typically assume constant factor loadings that cannot accommodate the state-dependent nature of production-based risks [Kogan and Papanikolaou \(2013\)](#); [Garlappi and Song \(2017\)](#).

The Residual Asset Burden strategy delivers economically and statistically significant returns across multiple specifications and sample periods. A value-weighted long/short trading strategy based on RAB achieves an annualized gross Sharpe ratio of 0.39 and a monthly average return of 24 basis points with a t-statistic of 3.06. The strategy's alpha remains robust across all standard factor models, ranging from 23 to 35 basis points per month, with the highest alpha of 35 basis points (t-statistic = 5.06) relative to the Fama-French three-factor model and the lowest alpha of 23 basis points (t-statistic = 2.93) relative to the CAPM. These results demonstrate that RAB captures a dimension of systematic risk not spanned by existing factors.

The economic significance of RAB persists after accounting for transaction costs and remains strong among large-capitalization stocks where implementation is most feasible. The net Sharpe ratio of 0.35 indicates that the strategy remains profitable after incorporating trading costs using high-frequency effective spread estimates. Among stocks with market capitalization above the 80th NYSE percentile, the RAB strategy generates an average return of 19 basis points per month (t-statistic = 2.17), with alphas ranging from 20 to 32 basis points relative to standard factor models. The strategy maintains statistical significance across alternative portfolio construction methodologies, with twenty-two out of twenty-five specifications yielding t-statistics exceeding two.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the RAB trading strategy achieves an alpha of 21 basis points per month with a t-statistic of 3.02. This result establishes that RAB contains unique information about expected returns beyond existing predictors in the factor zoo. The strategy loads significantly on the HML factor with a coefficient of -0.39 (t-statistic = -11.91), consistent with high RAB firms exhibiting growth-like characteristics despite their asset-intensive nature. The positive loading on RMW (0.19, t-statistic = 5.75) indicates that high RAB firms maintain robust profitability despite their production frictions, supporting the interpretation that these firms earn risk premia rather than exhibiting distress.

This paper contributes to the production-based asset pricing literature by introducing Residual Asset Burden as a novel measure of firms' exposure to productivity and adjustment cost shocks. While [Zhang \(2005a\)](#) establishes the theoretical link between investment frictions and expected returns, and [Liu et al. \(2009\)](#) demonstrate that investment-based factors explain cross-sectional return variation, our RAB measure provides a direct empirical proxy for the friction component of asset intensity that drives risk premia in production economies. Unlike the operating leverage mea-

sure of [Novy-Marx \(2011\)](#), which captures the sensitivity of earnings to sales, RAB isolates the systematic risk arising from inflexible productive capacity that cannot be easily redeployed in response to productivity shocks.

Our findings extend the investment-based explanation of the value premium documented in [Zhang \(2005b\)](#) and [Li and Zhang \(2010\)](#) by showing that residual asset burden captures a distinct source of systematic risk orthogonal to book-to-market ratios and other value measures. The RAB strategy achieves a monthly alpha of 23 basis points (t -statistic = 3.27) after controlling for book-to-market in Fama-MacBeth regressions, demonstrating that production frictions generate return predictability beyond traditional value effects. This result reconciles the seemingly contradictory findings that both asset-light and asset-heavy firms can earn premium returns by recognizing that it is the friction-adjusted burden of assets, rather than asset intensity per se, that determines exposure to systematic risk.

Methodologically, we establish RAB’s robustness using the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#), demonstrating that the anomaly survives multiple hurdles including transaction costs, alternative portfolio constructions, and controls for related factors. The strategy’s net Sharpe ratio of 0.35 places it in the 93rd percentile of the factor zoo, while its ability to expand the mean-variance frontier even after controlling for six factors and six related anomalies confirms its economic significance. These results provide empirical support for production-based asset pricing models and highlight the importance of incorporating real frictions into our understanding of cross-sectional return variation. The persistence of the RAB premium across market conditions and firm sizes suggests that markets efficiently price production-based risks, but these risks represent a fundamental source of systematic variation that cannot be diversified away.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial statement items that enable consistent measurement across firms and time periods. Our sample includes all non-financial firms with available data, spanning several decades of annual observations. Residual Asset Burden signal utilizes two key accounting variables from firms' balance sheets. Current assets other sundry (ACOX) represents miscellaneous current assets that do not fit into standard current asset categories such as cash, receivables, or inventory. This item captures various prepaid expenses, deferred charges, and other residual current assets that firms report but do not classify elsewhere. Stockholders' equity (SEQ) measures the book value of shareholders' ownership interest in the firm, calculated as total assets minus total liabilities. This fundamental accounting measure represents the net worth attributable to equity holders based on historical cost accounting principles. We construct the Residual Asset Burden signal by dividing current assets other sundry (ACOX) by stockholders' equity (SEQ) for each firm-year observation. This ratio captures the relative magnitude of miscellaneous current assets scaled by the firm's equity base, providing insight into the proportion of shareholders' equity tied up in sundry current assets that may represent lower-quality or less liquid resources. We calculate this measure using fiscal year-end values from annual financial statements. To ensure data quality, we require non-missing and positive values for stockholders' equity, as negative or zero equity would render the ratio economically meaningless. We also exclude observations with missing values for current assets other sundry, treating missing values as zero only when the firm reports other current asset categories, suggesting true absence rather than data unavailability.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RAB signal. Panel A plots the time-series of the mean, median, and interquartile range for RAB. On average, the cross-sectional mean (median) RAB is 0.06 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input RAB data. The signal's interquartile range spans -0.00 to 0.09. Panel B of Figure 1 plots the time-series of the coverage of the RAB signal for the CRSP universe. On average, the RAB signal is available for 6.23% of CRSP names, which on average make up 7.21% of total market capitalization.

4 Does RAB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RAB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RAB portfolio and sells the low RAB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RAB strategy earns an average return of 0.24% per month with a t-statistic of 3.06. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.39,

with a t-statistic of -11.91 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least \$1,551 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 12 bps/month with a t-statistics of 1.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 2-24bps/month. The lowest return, (2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.35. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RAB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the RAB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RAB, as well as average returns and alphas for long/short trading RAB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RAB strategy achieves an average return of 19 bps/month with a t-statistic of 2.17. Among these large cap stocks, the alphas for the RAB strategy relative to the five most common factor models range from 20 to 32 bps/month with t-statistics between 2.26 and 4.15.

5 How does RAB perform relative to the zoo?

Figure 2 puts the performance of RAB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RAB strategy falls in the distribution. The RAB strategy’s gross (net) Sharpe ratio of 0.39 (0.35) is greater than 81% (93%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RAB strategy (red line).² Ignoring trading costs, a \$1 invested in the RAB strategy would have yielded \$4.12 which ranks the RAB strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RAB strategy would have yielded \$3.24 which ranks the RAB strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RAB relative to those. Panel A shows that the RAB strategy gross alphas fall between the 49 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RAB strategy has a positive net generalized alpha for five out of the five factor models. In these cases RAB ranks between the 69 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does RAB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RAB with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RAB or at least to weaken the power RAB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RAB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RAB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RAB}RAB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RAB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RAB. Stocks are finally grouped into five RAB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

RAB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RAB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RAB signal in these Fama-MacBeth regressions exceed 2.15, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on RAB is 2.72.

Similarly, Table 5 reports results from spanning tests that regress returns to the RAB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RAB strategy earns alphas that range from 22-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.15, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RAB trading strategy achieves an alpha of 21bps/month with a t-statistic of 3.02.

7 Does RAB add relative to the whole zoo?

Finally, we can ask how much adding RAB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the RAB signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes RAB grows to \$2453.27.

8 Conclusion

This study provides robust evidence that Residual Asset Burden (RAB) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that RAB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of RAB as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.39 gross (0.35 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. More notably, the strategy delivers statistically significant abnormal returns of 25 basis points per month (t -statistic = 3.58) relative to the Fama-French five-factor model augmented with momentum, suggesting that RAB captures a distinct source of systematic risk or mispricing not explained by conventional factors.

The robustness of our results is particularly striking when considering the strategy’s performance against a comprehensive set of related factors. Even after control-

ization on CRSP in the period for which RAB is available.

ling for seven closely related strategies including traditional value measures, leverage metrics, and earnings-based signals, RAB maintains a significant alpha of 21 basis points per month (t -statistic = 3.02). This finding suggests that RAB contains unique information content about future returns that is not subsumed by existing factors in the literature.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The persistence of RAB’s predictive power after transaction costs indicates its potential utility in real-world portfolio construction. The signal’s orthogonality to established factors suggests it could serve as a valuable addition to multi-factor models and quantitative investment strategies.

Despite these compelling findings, several limitations warrant acknowledgment. First, while we employ state-of-the-art methodology to address data-mining concerns, the possibility of overfitting in the broader factor zoo remains a consideration. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of RAB’s predictive power to international markets requires further investigation. Third, the economic mechanisms underlying RAB’s return predictability remain somewhat opaque, calling for theoretical development to better understand the sources of these returns.

Future research should address several promising avenues. First, examining RAB’s performance across different market regimes and economic cycles would provide insights into its stability and conditional effectiveness. Second, investigating the signal’s behavior in international markets would test its robustness across different institutional and regulatory environments. Third, developing theoretical models that explain why firms with higher residual asset burdens generate lower future returns would enhance our understanding of the underlying economic forces. Finally, exploring potential interactions between RAB and other firm characteristics, particularly in the context of corporate financial policies and investment decisions, could yield

valuable insights for both asset pricing and corporate finance literature.

In conclusion, this paper establishes Residual Asset Burden as a robust and economically significant predictor of stock returns that warrants inclusion in the canon of documented asset pricing anomalies. Its strong performance, both in isolation and after controlling for related factors, suggests that RAB captures an important dimension of the cross-section of expected returns that has been previously overlooked in the literature.

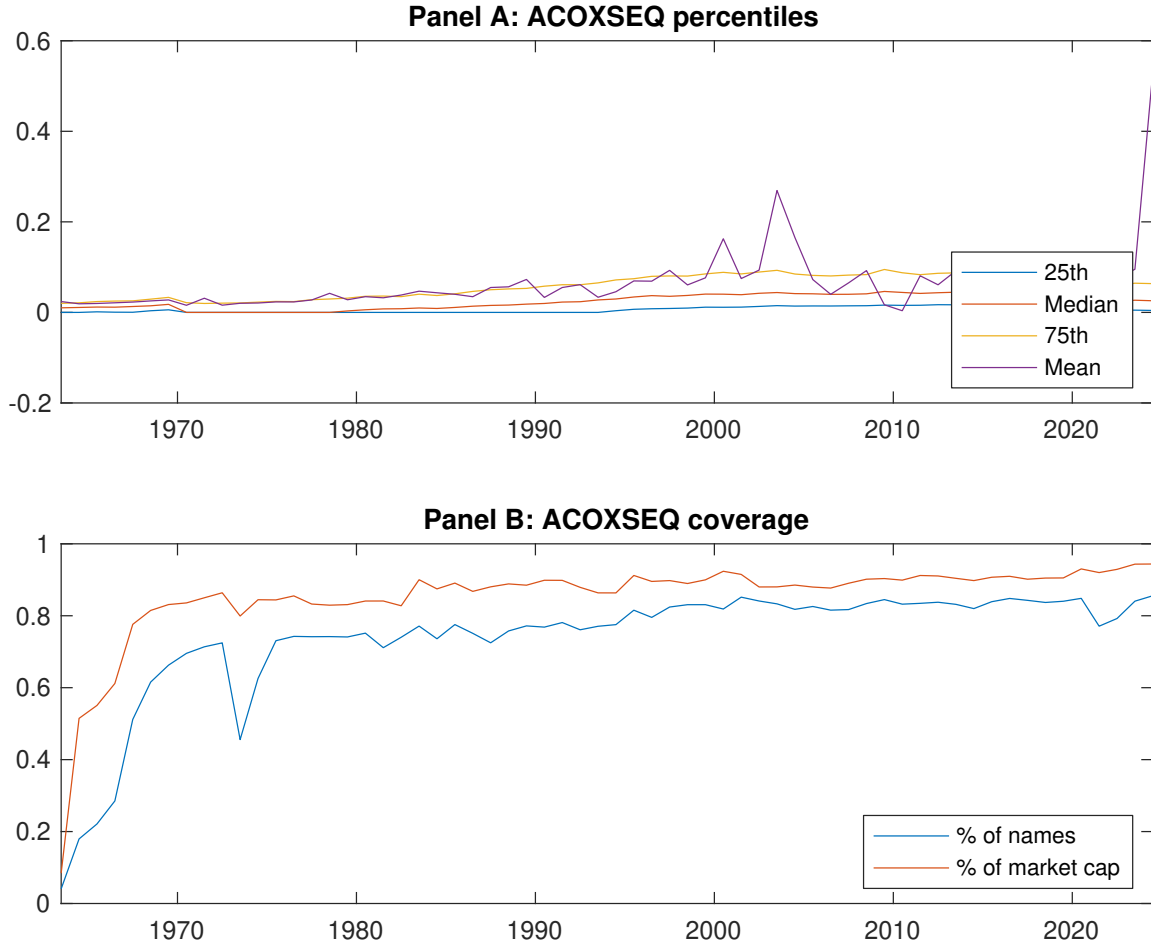


Figure 1: Times series of RAB percentiles and coverage. This figure plots descriptive statistics for RAB. Panel A shows cross-sectional percentiles of RAB over the sample. Panel B plots the monthly coverage of RAB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RAB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on RAB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.53 [3.01]	0.55 [3.04]	0.61 [3.58]	0.62 [3.65]	0.77 [4.38]	0.24 [3.06]
α_{CAPM}	-0.07 [-1.26]	-0.07 [-1.27]	0.02 [0.55]	0.04 [0.81]	0.17 [3.52]	0.23 [2.93]
α_{FF3}	-0.14 [-3.20]	-0.05 [-0.90]	0.05 [1.09]	0.09 [2.06]	0.21 [4.46]	0.35 [5.06]
α_{FF4}	-0.12 [-2.71]	-0.02 [-0.29]	0.02 [0.57]	0.06 [1.46]	0.20 [4.13]	0.32 [4.53]
α_{FF5}	-0.17 [-3.62]	0.01 [0.24]	0.03 [0.59]	0.07 [1.66]	0.10 [2.40]	0.27 [3.92]
α_{FF6}	-0.15 [-3.18]	0.04 [0.66]	0.01 [0.20]	0.05 [1.18]	0.10 [2.33]	0.25 [3.58]
Panel B: Fama and French (2018) 6-factor model loadings for RAB-sorted portfolios						
β_{MKT}	1.06 [93.77]	1.02 [79.40]	0.99 [92.22]	0.98 [94.45]	1.02 [96.25]	-0.04 [-2.19]
β_{SMB}	-0.08 [-5.13]	-0.01 [-0.65]	0.05 [3.51]	0.00 [0.01]	0.08 [5.49]	0.17 [6.86]
β_{HML}	0.21 [9.73]	-0.04 [-1.67]	-0.08 [-3.88]	-0.12 [-5.80]	-0.18 [-8.65]	-0.39 [-11.91]
β_{RMW}	0.05 [2.10]	-0.13 [-5.26]	0.04 [2.01]	0.05 [2.50]	0.24 [11.41]	0.19 [5.75]
β_{CMA}	0.05 [1.49]	-0.06 [-1.53]	0.02 [0.79]	-0.03 [-0.99]	0.11 [3.81]	0.07 [1.39]
β_{UMD}	-0.03 [-2.42]	-0.03 [-2.55]	0.02 [2.33]	0.03 [2.79]	0.00 [0.21]	0.03 [1.75]
Panel C: Average number of firms (n) and market capitalization (me)						
n	683	813	705	598	559	
me (\$10 ⁶)	1551	1611	1899	2218	2197	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RAB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.24 [3.06]	0.23 [2.93]	0.35 [5.06]	0.32 [4.53]	0.27 [3.92]	0.25 [3.58]
Quintile	NYSE	EW	0.20 [3.66]	0.15 [2.89]	0.14 [2.74]	0.14 [2.72]	0.09 [1.76]	0.10 [1.84]
Quintile	Name	VW	0.26 [3.34]	0.26 [3.22]	0.38 [5.42]	0.33 [4.73]	0.30 [4.26]	0.27 [3.78]
Quintile	Cap	VW	0.19 [2.56]	0.19 [2.56]	0.30 [4.62]	0.27 [4.04]	0.25 [3.68]	0.22 [3.28]
Decile	NYSE	VW	0.12 [1.35]	0.11 [1.16]	0.22 [2.60]	0.21 [2.50]	0.21 [2.41]	0.20 [2.34]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.22 [2.73]	0.21 [2.63]	0.32 [4.49]	0.30 [4.22]	0.24 [3.42]	0.22 [3.24]
Quintile	NYSE	EW	0.02 [0.35]					
Quintile	Name	VW	0.24 [3.01]	0.24 [2.94]	0.34 [4.86]	0.32 [4.51]	0.26 [3.80]	0.24 [3.55]
Quintile	Cap	VW	0.16 [2.21]	0.16 [2.11]	0.26 [3.91]	0.24 [3.60]	0.20 [3.01]	0.19 [2.80]
Decile	NYSE	VW	0.09 [0.95]	0.07 [0.76]	0.17 [2.01]	0.17 [1.97]	0.14 [1.68]	0.14 [1.65]

Table 3: Conditional sort on size and RAB

This table presents results for conditional double sorts on size and RAB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RAB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RAB and short stocks with low RAB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RAB Quintiles					RAB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.65 [2.66]	0.51 [1.97]	0.83 [3.26]	0.85 [3.38]	0.86 [3.07]	0.20 [1.80]	0.15 [1.33]	0.11 [0.95]	0.09 [0.75]	0.06 [0.51]	0.05 [0.39]
	(2)	0.74 [3.16]	0.74 [3.12]	0.74 [3.10]	0.72 [3.03]	0.92 [3.78]	0.18 [2.17]	0.15 [1.80]	0.13 [1.54]	0.17 [2.06]	0.09 [1.01]	0.13 [1.49]
	(3)	0.70 [3.33]	0.67 [3.14]	0.76 [3.50]	0.88 [3.92]	0.95 [4.19]	0.25 [2.85]	0.20 [2.27]	0.19 [2.19]	0.18 [1.96]	0.17 [1.90]	0.16 [1.73]
	(4)	0.62 [3.10]	0.56 [2.84]	0.72 [3.57]	0.87 [4.28]	0.79 [3.83]	0.17 [1.99]	0.15 [1.75]	0.22 [2.59]	0.21 [2.41]	0.15 [1.77]	0.15 [1.72]
	(5)	0.52 [3.01]	0.57 [3.23]	0.52 [3.24]	0.56 [3.36]	0.71 [4.22]	0.19 [2.17]	0.20 [2.26]	0.32 [4.15]	0.29 [3.59]	0.24 [3.09]	0.22 [2.72]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RAB Quintiles					RAB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	377	385	386	386	378	29	31	33	32	29	
	(2)	103	103	104	103	103	53	54	54	54	54	
	(3)	72	72	72	72	72	92	90	92	92	94	
	(4)	60	59	60	59	60	199	193	196	197	204	
(5)	54	54	54	54	54	1344	1401	1461	1788	1611		

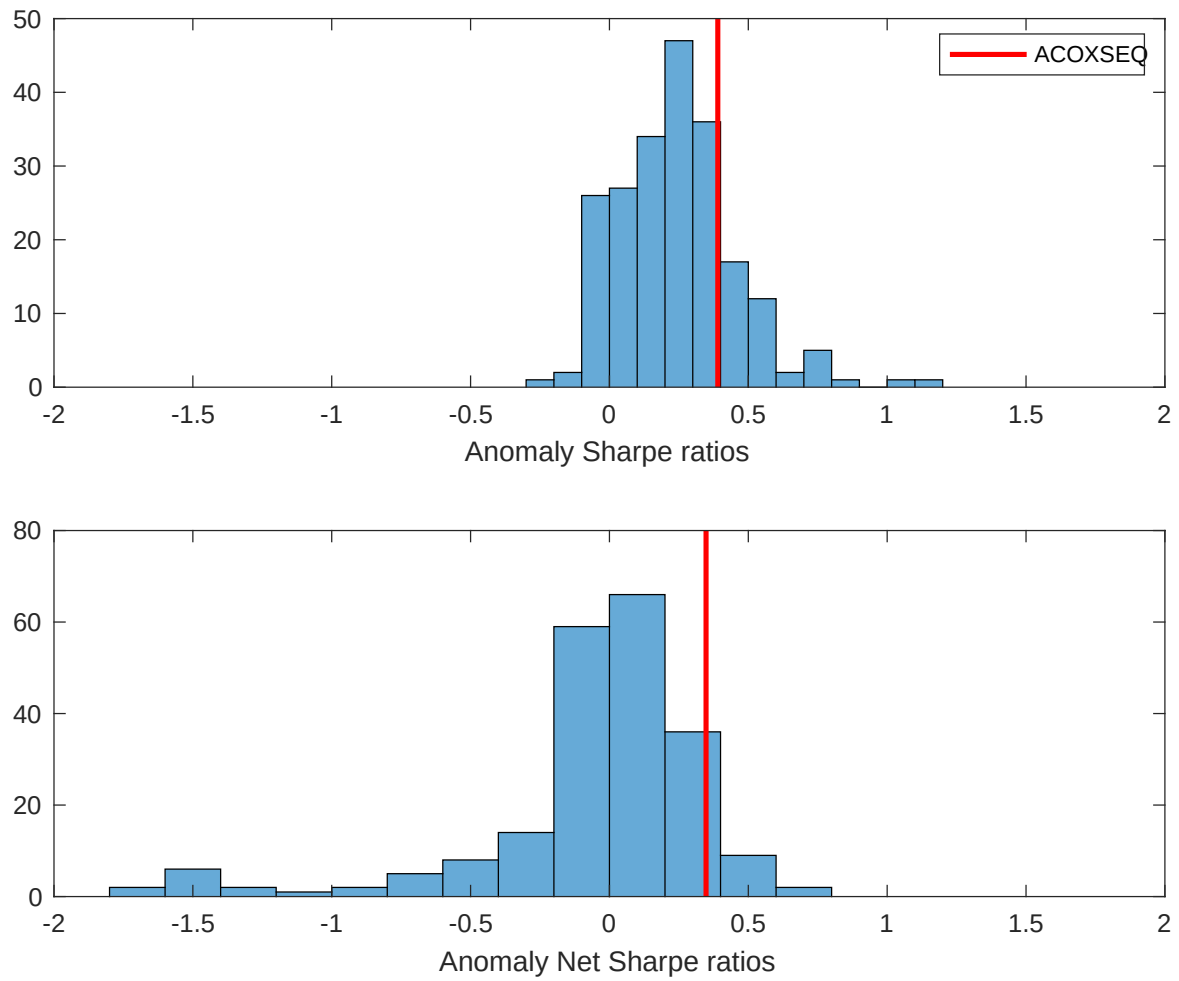


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RAB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

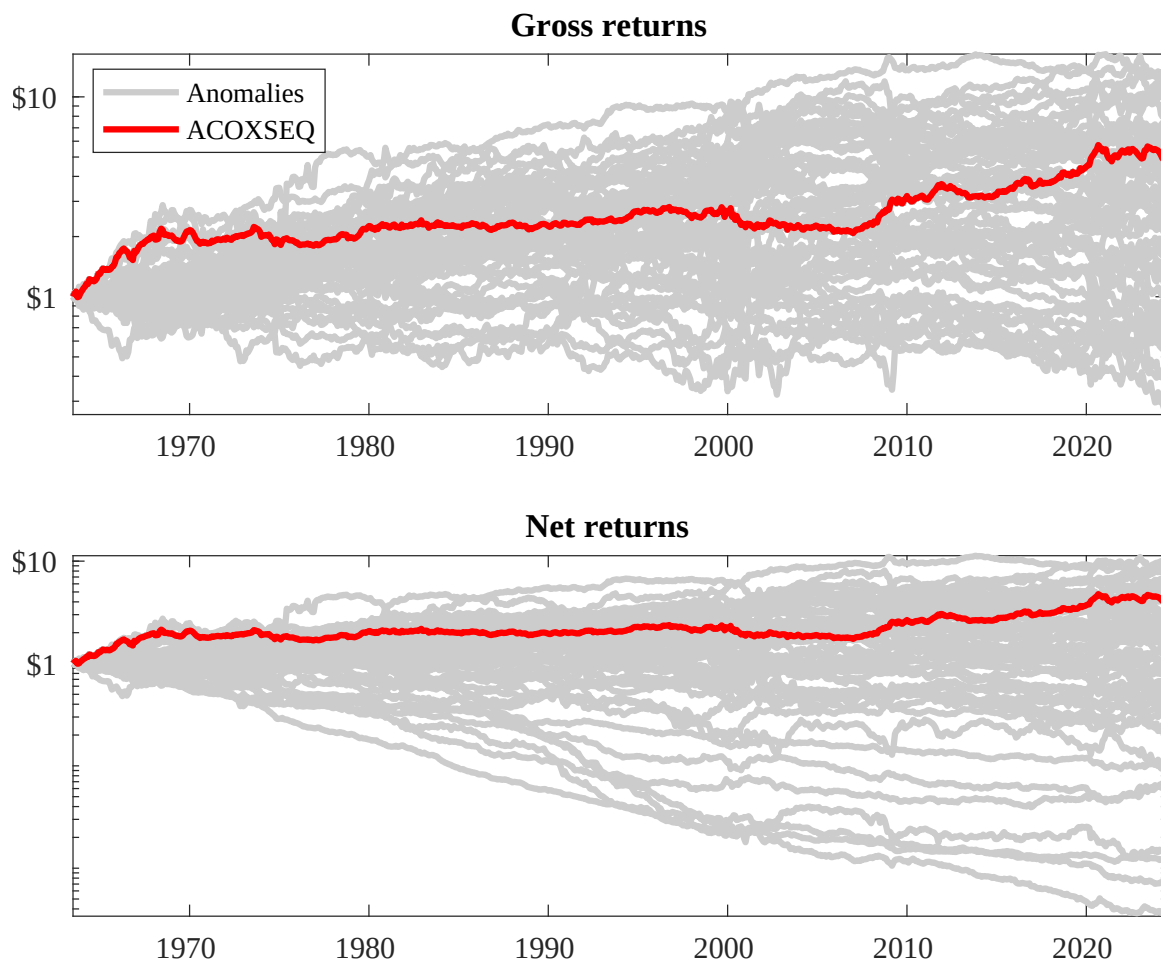
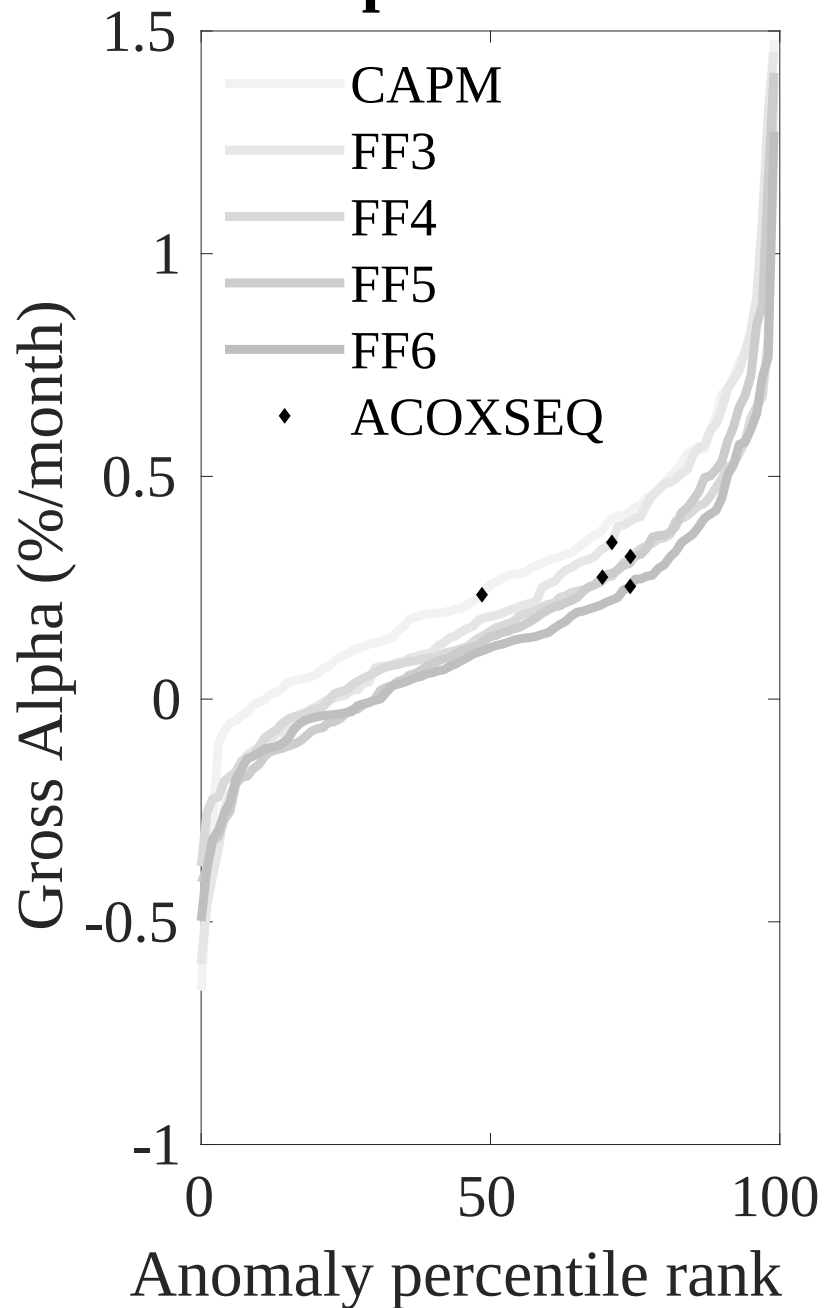


Figure 3: Dollar invested.

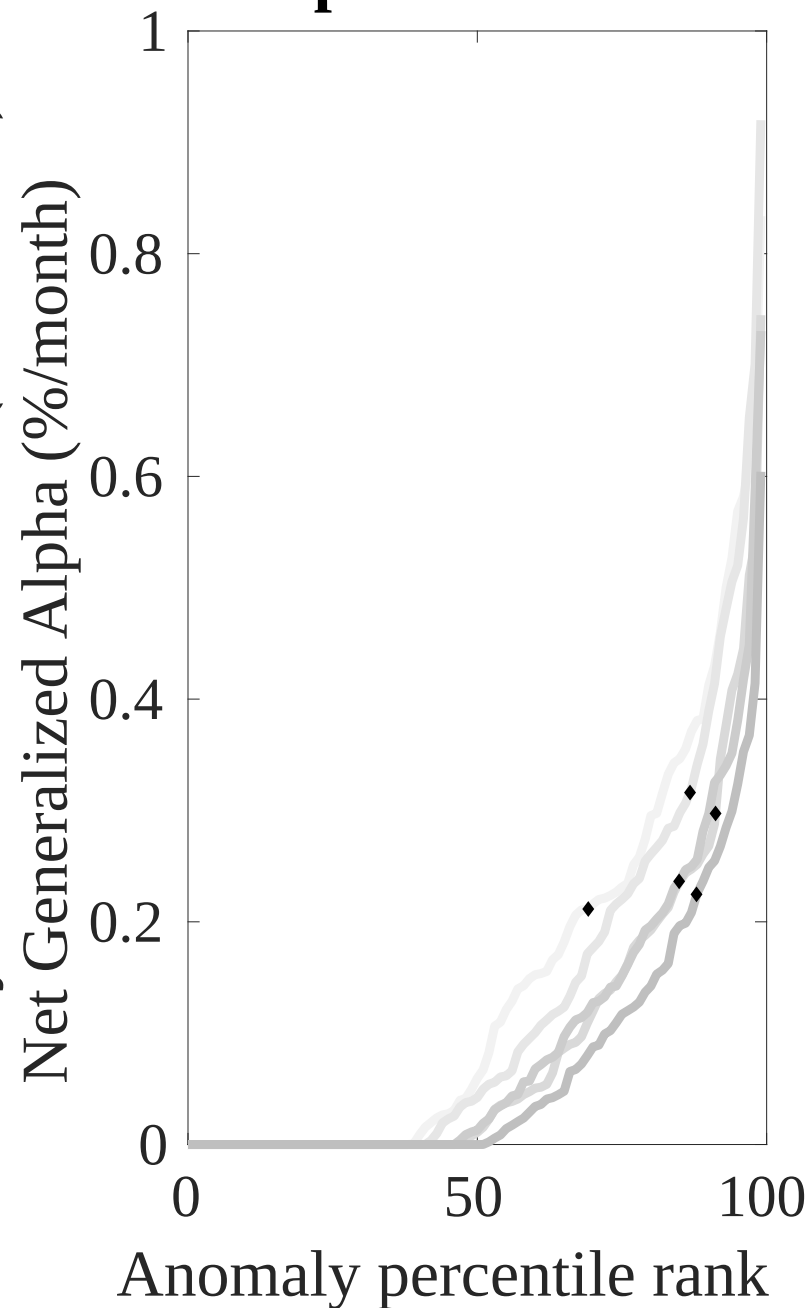
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RAB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



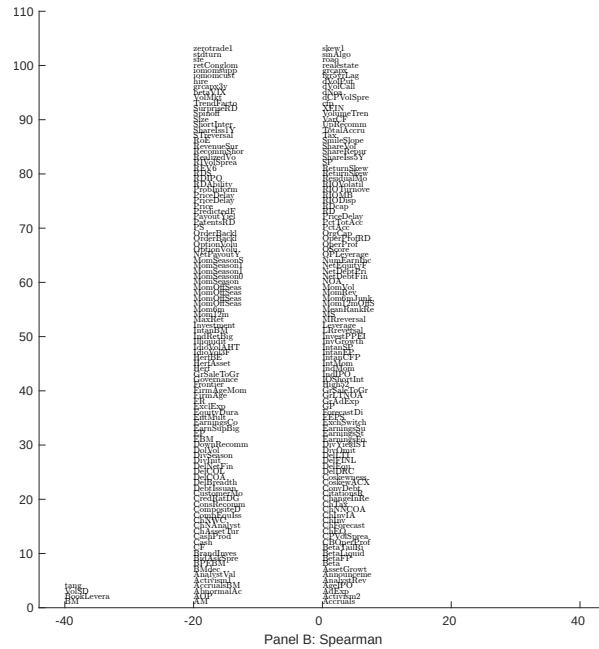
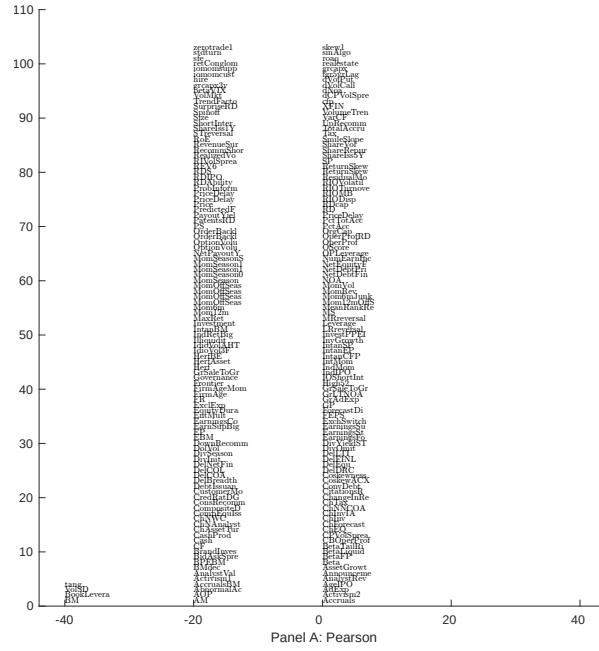


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with RAB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

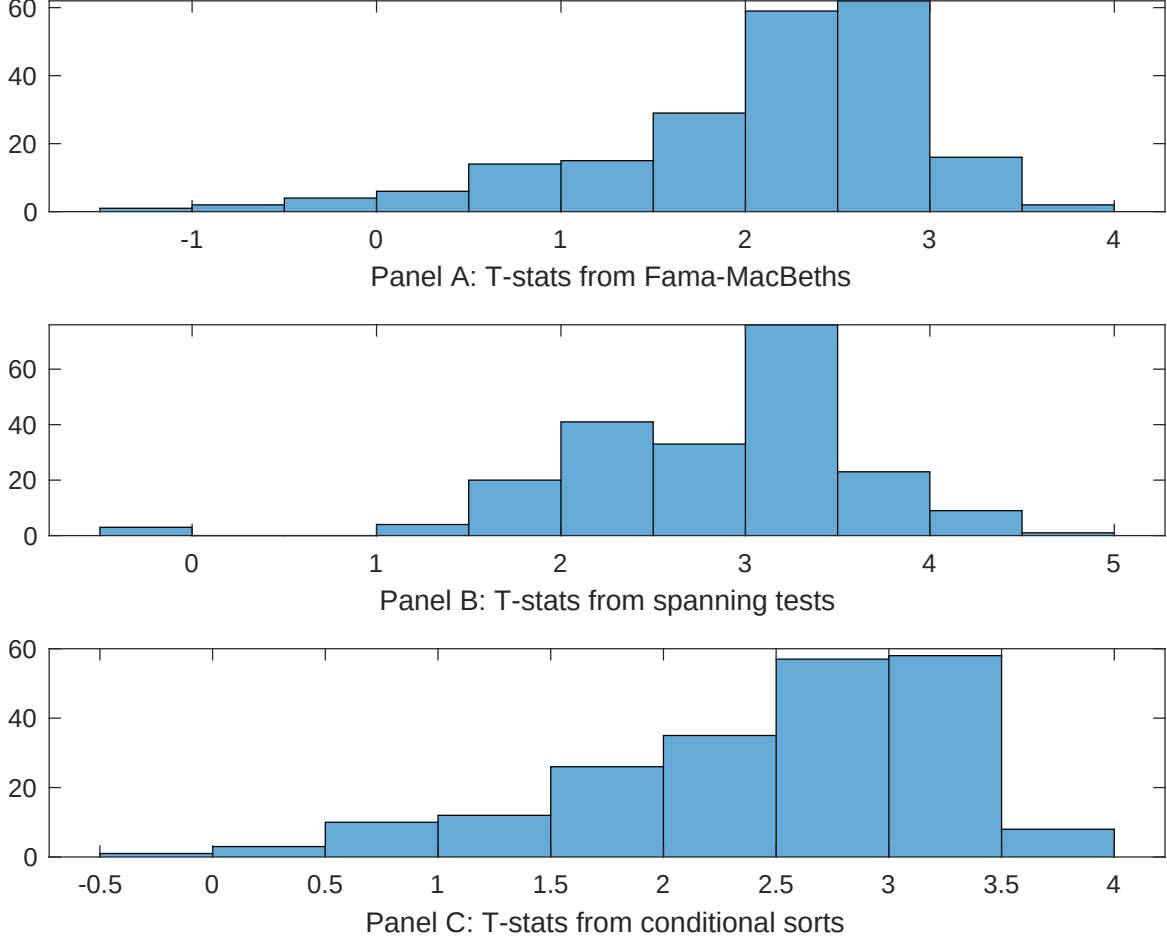


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RAB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RAB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RAB}RAB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RAB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RAB. Stocks are finally grouped into five RAB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RAB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RAB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RAB}RAB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.18]	0.12 [5.06]	0.10 [4.34]	0.20 [9.00]	0.11 [4.83]	0.13 [6.17]	0.21 [8.49]
RAB	0.13 [3.43]	0.11 [3.34]	0.74 [2.15]	0.12 [3.51]	0.81 [2.46]	0.87 [2.53]	0.90 [2.72]
Anomaly 1	0.37 [7.52]						0.87 [1.95]
Anomaly 2		0.13 [1.27]					-0.18 [-1.23]
Anomaly 3			0.12 [3.31]				0.88 [2.69]
Anomaly 4				0.51 [5.30]			0.60 [5.61]
Anomaly 5					0.17 [2.57]		-0.59 [-0.84]
Anomaly 6						0.79 [3.41]	0.10 [3.21]
# months	715	720	720	720	715	715	715
$\bar{R}^2(\%)$	1	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RAB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RAB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Book to market, original (Statman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.23 [3.27]	0.22 [3.15]	0.23 [3.27]	0.25 [3.47]	0.26 [3.60]	0.30 [4.14]	0.21 [3.02]
Anomaly 1	-23.37 [-5.76]						-19.40 [-4.19]
Anomaly 2		10.55 [4.02]					7.48 [2.67]
Anomaly 3			19.47 [7.03]				16.55 [5.73]
Anomaly 4				-11.77 [-3.23]			-2.92 [-0.66]
Anomaly 5					-5.53 [-1.76]		11.04 [2.78]
Anomaly 6						-11.31 [-3.91]	-9.06 [-2.64]
mkt	-2.75 [-1.64]	-2.24 [-1.30]	-3.40 [-2.06]	-3.45 [-2.04]	-3.75 [-2.19]	-4.00 [-2.36]	-1.62 [-0.97]
smb	25.75 [9.06]	16.46 [6.75]	8.35 [3.11]	20.48 [7.70]	18.23 [7.16]	18.98 [7.62]	16.52 [5.35]
hml	-14.38 [-2.75]	-31.20 [-8.41]	-28.09 [-8.02]	-25.17 [-4.82]	-32.87 [-7.40]	-30.24 [-7.88]	-5.49 [-0.96]
rmw	15.25 [4.60]	22.01 [6.51]	8.09 [2.28]	18.52 [5.61]	19.67 [5.85]	19.95 [6.03]	8.15 [2.10]
cma	8.22 [1.73]	6.35 [1.33]	2.57 [0.54]	4.00 [0.83]	5.79 [1.19]	6.91 [1.44]	5.79 [1.23]
umd	4.22 [2.54]	2.67 [1.59]	2.74 [1.68]	3.00 [1.79]	2.20 [1.24]	-0.86 [-0.44]	2.07 [1.03]
# months	728	732	732	732	728	728	728
$\bar{R}^2(\%)$	31	30	33	29	28	30	36

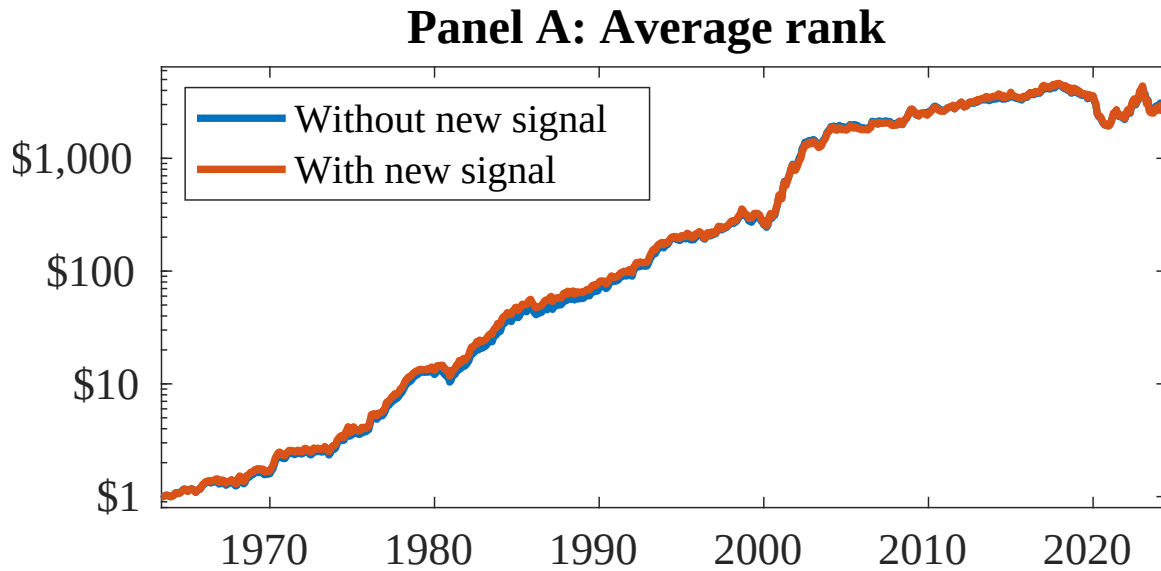


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as RAB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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